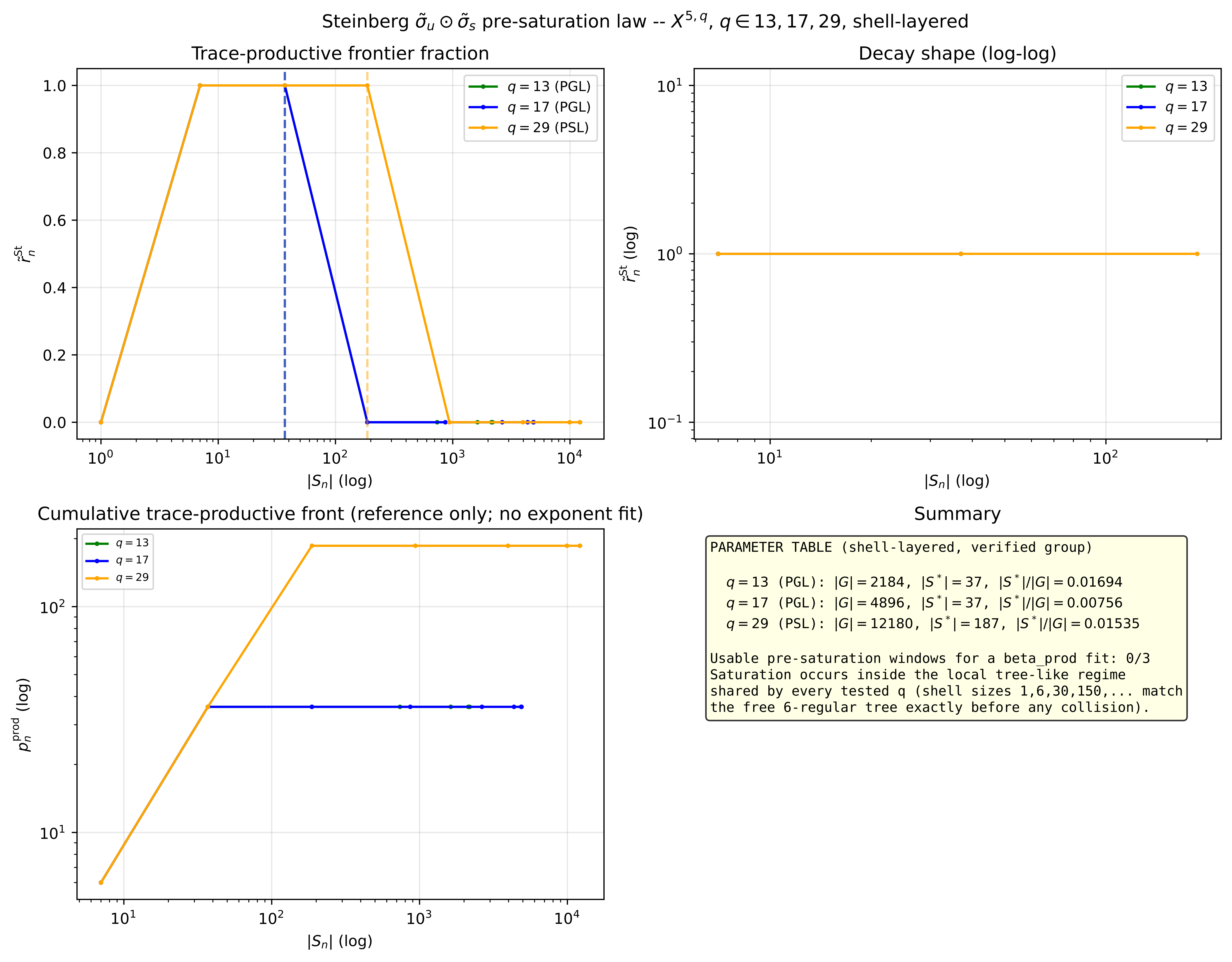
Steinberg $\tilde{\sigma}_u \odot \tilde{\sigma}_s$ pre-saturation law -- $X^{5,q}$, $q \in 13, 17, 29$, shell-layered
Trace-productive frontier fraction
$\tilde{r}_n^{\mathrm{St}}$
$|S_n|$ (log)
$q = 13$ (PGL)
$q = 17$ (PGL)
$q = 29$ (PSL)
Decay shape (log-log)
$\tilde{r}_n^{\mathrm{St}}$ (log)
$|S_n|$ (log)
$q = 13$
$q = 17$
$q = 29$
Cumulative trace-productive front (reference only; no exponent fit)
p_n^{prod} (log)
$|S_n|$ (log)
$q = 13$
$q = 17$
$q = 29$
Summary
PARAMETER TABLE (shell-layered, verified group)
$q = 13$ (PGL): $|G| = 2184$, $|S^*| = 37$, $|S^*|/|G| = 0.01694$
$q = 17$ (PGL): $|G| = 4896$, $|S^*| = 37$, $|S^*|/|G| = 0.00756$
$q = 29$ (PSL): $|G| = 12180$, $|S^*| = 187$, $|S^*|/|G| = 0.01535$
Usable pre-saturation windows for a beta_prod fit: 0/3
Saturation occurs inside the local tree-like regime
shared by every tested q (shell sizes 1,6,30,150,... match
the free 6-regular tree exactly before any collision).